

Fake News Detection by Fine-Tuning a BERT-style Language Model

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1.Introduction

Automatic fake news detection is a type of text classification where the goal is to determine if a news article is fake or real. This is important because false information spreads quickly online, and it's hard for people to check every story manually. The best methods today use special computer models called pre-trained transformer language models. These models learn general language patterns from a lot of text without labels and then are adjusted, or fine-tuned, to work on a specific task using a smaller set of labelled examples (Devlin et al., 2019; Radford et al., 2018). This project uses a smaller version of a popular model called Distil BERT, which is a lighter version of BERT (Sanh et al., 2019). To understand how fine-tuning helps, we compare it with older methods that look at word frequencies and with the same transformer model without any fine-tuning.

2.Dataset

We use a dataset called GonzaloA/fake news from Hugging Face Hub, which includes about 45,000 English news articles. Each article is labelled as true (real) or false (fake) and has a title and full text. The original data was split into train, validation, and test sets, but we combined them and made a new split with 80% for training and 20% for testing. This way, every model is trained and tested on the same set of articles, making the comparison fair. To make the work easier and fit on a free Google Collab T4 GPU, we reduced the number of articles to 8,000 for training and 2,000 for testing. The classes are almost balanced. One thing to note about this dataset is that the real articles mostly come from Reuters news stories, which have a unique writing style we talk more about this in the discussion section.

3.Model and Methodology

Preprocessing: The text was split into smaller parts using Distil Bert's Word Piece tokenizer. This splits words into smaller units, adds special tokens like [CLS] and [SEP], makes sure each article has exactly 256 tokens by either shortening or adding padding, and creates an attention mask to show which tokens are important.

Models compared:

- I. Two traditional models: TF-IDF features using single words and pairs of words (unigrams and bigrams) with a vocabulary of 20,000, combined with Logistic Regression and Multinomial Naive Bayes.
- II. A baseline without fine-tuning: a pre-trained Distil BERT model that is not updated, used only to extract features from the [CLS] token, then combined with Logistic Regression.
- III. The fine-tuned model: a pre-trained Distil BERT with about 66 million parameters, which has a classification layer for two classes and is trained end-to-end. This is full fine-tuning, not using adapters or prompts.

Training: The fine-tuned model was trained using a PyTorch loop with the AdamW optimizer (learning rate of 2×10^{-5}), calculating loss based on the model's predictions, with a linear learning rate schedule, gradient clipping set to 1.0, a batch size of 16, and three training epochs. The test set was checked after each epoch to see how the model was learning, and all random settings were fixed to ensure results can be reproduced. We reported accuracy, as well as macro-averaged precision, recall, and F1 scores, treating both classes equally, along with a confusion matrix.

4.Results and Discussion

Table 1 shows the four models. The fine-tuned Distil BERT model performed the best, with 98.85% accuracy and a macro-F1 score of 0.988. It only incorrectly classified 23 out of 2,000 test articles (Figure 2: 12 fake articles were labelled as real, and 11 real articles were labelled as fake). The learning curve (Figure 1) explains this clearly: before training, the model guessed randomly and scored around 51.3%, which is like chance. Then, after just one training round, the model's performance jumped to 98.3% as the loss dropped from 0.07 to 0.01. Accuracy improved just a little more in the next two rounds (98.7%, then 98.85%) while the loss continued to decrease this shows that the model is still improving but at a slower rate, and there's no sign of overfitting after three rounds of training.

Table 1: Test-set performance of all four models

Model	Accuracy	Precision	Recall	F1
TF-IDF+Logistic Regression	0.9640	0.9642	0.9633	0.9637
TFI-IDF+Navie Bayes	0.9335	0.932	0.9330	0.9331
Frozen DistillBERT features +LogReg(no fine-tuning)	0.9750	0.9750	0.9747	0.9748
DistillBERT (fine-tuned)	0.9885	0.9885	0.9884	0.9884

The comparison also helps show how much fine-tuning really matters. The traditional approach using TF-IDF and Logistic Regression already got 96.4% accuracy. Even without any training, using features from a frozen Distil BERT model got 97.5%. Fine-tuning helped a little, but only by about one or two percentage points. The main point here isn't that BERT "wins," but that this task was surprisingly simple. Because the dataset mostly has Reuters news articles that are well-structured and use similar words, even a simple model that just looks at words can tell the classes apart based on surface level clues, not real understanding of truth. The high accuracy here comes from the dataset itself and doesn't mean the model can handle fake news from different sources, topics, or times something that's well known to be a problem with this kind of data.

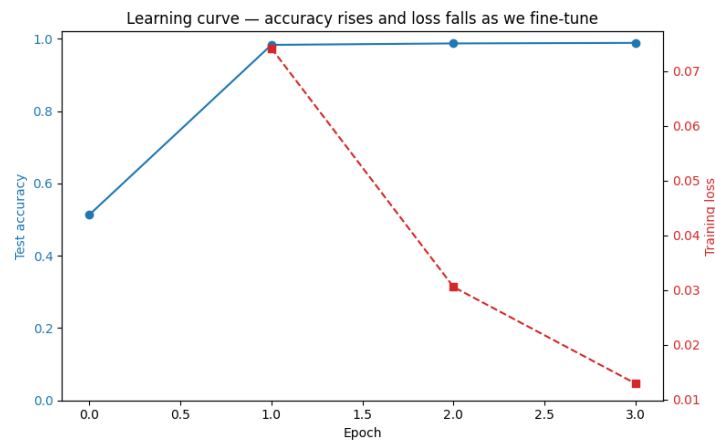


Figure 1: Learning curve. Test accuracy (blue) rises from chance to 98% after one epoch while training loss (red) falls steadily

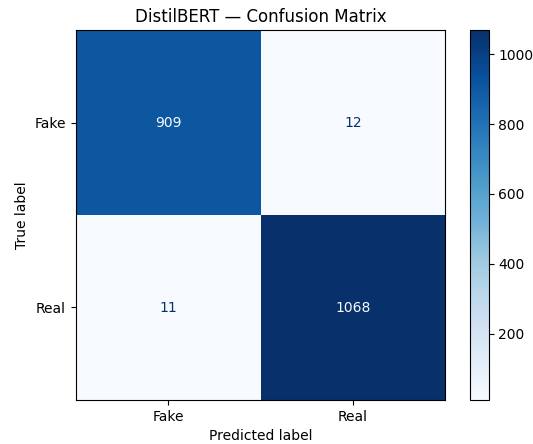


Figure 2: Confusion matrix for the fine-tuned model on the 2,000-article test set (23 total errors)

Limitations: The results are based on a single stratified split using one random seed. A sub-sample of the data (~8,000 for training and 2,000 for testing) was selected to fit within the free-GPU compute budget. Since we combined the original dataset splits and did not keep a separate validation set, we used the test set for monitoring during each training epoch this is a mild form of test-set peeking. Using a separate validation set would be better. We didn't do cross-validation or tune hyperparameters, and all evaluation was done only within the same domain.

5. Conclusion

Fine-tuning Distil BERT created a strong fake news classifier with 98.85% accuracy and a macro-F1 score of 0.988, which performed better than both the TF-IDF methods and the Distil BERT model without fine-tuning. However, the small advantage over a simple baseline, along with issues specific to the dataset's source, suggests that the high accuracy on this benchmark might not reflect the real-world challenges of the problem. Future research should test how well the model works across different sources, like the more difficult LIAR dataset, use k-fold cross-validation, try more efficient ways to fine-tune models like LoRa or adapters, and compare performance with larger models such as Bert-base or Roberta-base.

6. References

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